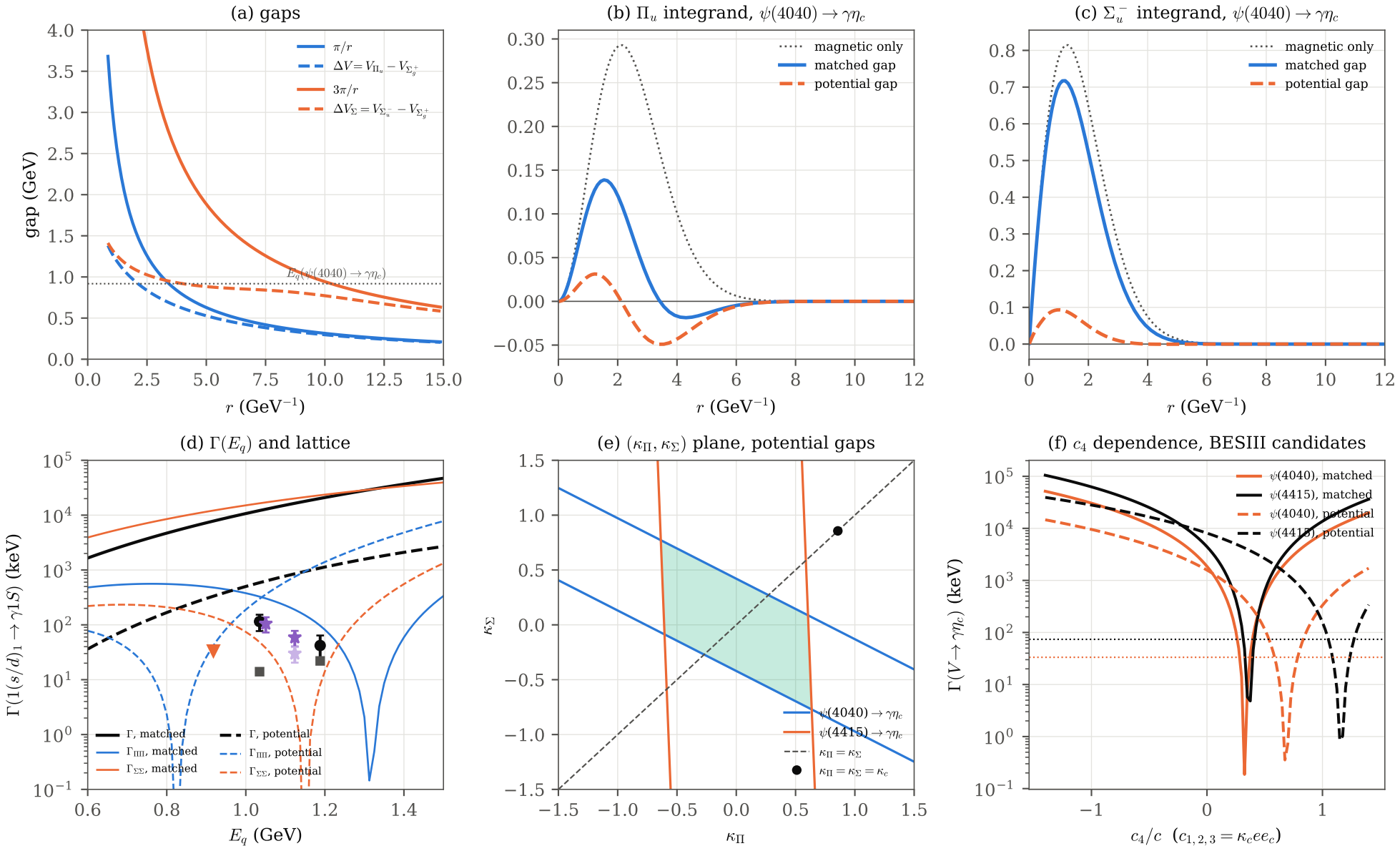

(a) gaps
gap (GeV)
r (GeV^{-1})
π/r
$\Delta V = V_{\Pi_u} - V_{\Sigma_g^+}$
$3\pi/r$
$\Delta V_\Sigma = V_{\Sigma_u^-} - V_{\Sigma_g^+}$
$E_q(\psi(4040) \to \gamma\eta_c)$
(b) Π_u integrand, $\psi(4040) \to \gamma\eta_c$
magnetic only
matched gap
potential gap
r (GeV^{-1})
(c) Σ_u^- integrand, $\psi(4040) \to \gamma\eta_c$
magnetic only
matched gap
potential gap
r (GeV^{-1})
(d) $\Gamma(E_q)$ and lattice
$\Gamma(1(s/d)_1 \to \gamma 1S)$ (keV)
Γ, matched
$\Gamma_{\Pi\Pi}$, matched
$\Gamma_{\Sigma\Sigma}$, matched
Γ, potential
$\Gamma_{\Pi\Pi}$, potential
$\Gamma_{\Sigma\Sigma}$, potential
E_q (GeV)
(e) $(\kappa_\Pi, \kappa_\Sigma)$ plane, potential gaps
κ_Σ
$\psi(4040) \to \gamma\eta_c$
$\psi(4415) \to \gamma\eta_c$
$\kappa_\Pi = \kappa_\Sigma$
$\kappa_\Pi = \kappa_\Sigma = \kappa_c$
κ_Π
(f) c_4 dependence, BESIII candidates
$\psi(4040)$, matched
$\psi(4415)$, matched
$\psi(4040)$, potential
$\psi(4415)$, potential
$\Gamma(V \to \gamma\eta_c)$ (keV)
c_4/c $(c_{1,2,3} = \kappa_c e e_c)$